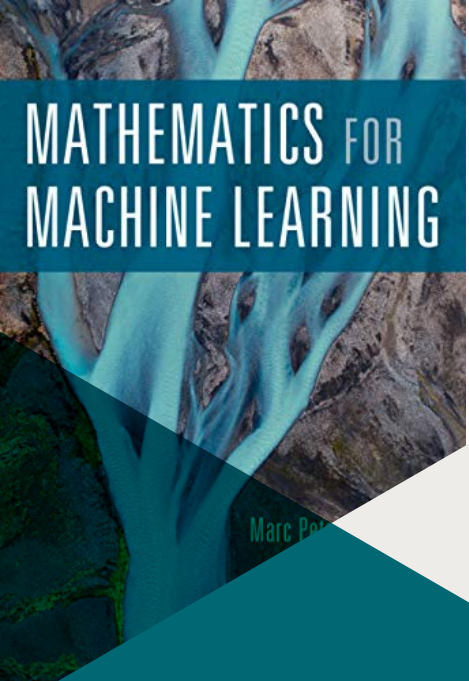

DATA SCIENCE UNINA

DMB - Machine Learning

6.2 - Activation, losses and metrics

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Overview

1. Introduction

Introduction

AI vs TML vs Deep Learning

- Understanding the differences between AI, Traditional Machine Learning (TML) and deep learning is crucial for practitioners and researchers.
- Each approach has its strengths and weaknesses, and the choice between them can significantly affect the outcome of a project.
- You must be able to select the most appropriate techniques for a specific applications, optimizing performance, efficiency, and resource utilization.

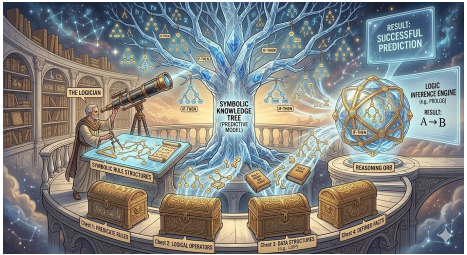
- **Definition of Symbolic AI**: it relies on high-level, human-readable symbols and formal logic to represent knowledge and solve problems. It operates on the principle that intelligence can be achieved through the explicit manipulation of rules, where "if-then" statements and structured databases define how the system reasons. Unlike data-driven models, it is a top-down approach where humans must manually codify every rule and relationship into the system.
- **Definition of Traditional Machine Learning**: a subset of artificial intelligence that focuses onthe development of algorithms and statistical models that enable computers to perform tasks without explicit instructions. Instead, ML systems learn from data, identifying patterns and making predictions or decisions based on that data. Key types of machine learning include supervised learning, unsupervised learning, and reinforcement learning.
- **Definition of Deep Learning**: is a specialized branch of machine learning that employs neural networks with many layers (hence "deep") to analyze various forms of data, particularly large datasets. Unlike traditional ML, which often requires manual feature extraction, deep learning models automatically learn feature representations from raw data. This allows them to excel in complex tasks such as image and speech recognition, natural language processing, and more.

Symbolic AI

Traditional AI systems are developed by analyzing the system and creating rules (like if... then) around it.

In my opinion it is closer to logic rather than anything else.

- The rules are explicitly programmed, significantly boosting the transparency of the system.
- These systems are also computationally efficient since there is no expensive training process.
- However, since these rules must be defined before running the system, traditional AI is often hampered in the scope of adaptation.
- In very complex (or high-dimensional systems) inferring rules/relationships between variables can be either impractical or impossible.



Traditional Machine Learning - TML

In my opinion, nowadays it is a subfield of statistics.

TML refers to a range of algorithms and techniques that allow computers to learn from data and make predictions or decisions without being explicitly programmed.

- **Feature Engineering**: requires human experts to select, modify, or create features from raw data which are used to improve model performance. It involves also transforming input data into formats that machine learning algorithms can better understand.
- **Data Size**: works well with small or medium datasets.
- **Hardware**: can run on standard CPUs.
- **Performances**: good not optimal.
- **Complexity and interpretability**: relatively low complexity and interpretability possible.

main applications

- Finance: For credit scoring, fraud detection, and algorithmic trading.
- Healthcare: In diagnosing diseases, predicting patient outcomes, andoptimizing treatment plans.
- Marketing: For customer segmentation, churn prediction, and personalized recommendations.
- Manufacturing: predictive maintenance and quality control. C.

Limitations

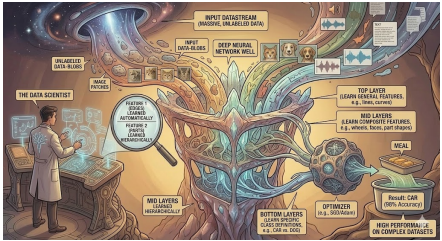
Despite its strengths, traditional machine learning has several limitations

- **Dependency on Feature Extraction**. TML models often require extensive manual feature extraction and selection. This dependency can be time-consuming and may require domain expertise to identify the most relevant features for a given problem.
- **Less Effective with Unstructured Data**. TML struggles with unstructured data types, such as images, audio, or text. While it can process structured data (like tables), it is less effective than DL in extracting Meaningful patterns from unstructured data without significant preprocessing and feature engineering.

Main algorithms:

- Decision Trees: a model that splits data into branches based on feature conditions, making decisions based on path outcomes.
- Support Vector Machines (SVMs): A classification technique that finds the hyperplane that best separates different classes in the feature space.
- Linear Regression: A method used for predicting a continuous outcome based on the linear relationship between input features.
- K-Nearest Neighbors (KNN): A classification technique that assigns a class based on the majority class of the nearest neighbors in the feature space.
- Random Forests: An ensemble method that builds multiple decision trees and merges their results to improve accuracy and control overfitting

Deep Learning - DL



In my opinion the most relevant field nowadays

Deep Learning (DL) is a specialized area of machine learning that uses neu networks with multiple layers (deep neural networks) to model complexpatterns in large datasets. This approach allows for the automatic learning of features directly from the raw data, leading to significant advancements in various fields.

- **Automatic Feature Extraction**. DL models automatically learn how to extract relevant features from raw input data. This reduces the need for manual feature engineering, allowing the model to discover intricate patterns and representations that may not be easily discernible by humans.

main applications

- **Image Recognition**: Used in facial recognition, object detection, andimageclassification tasks.
- **Speech Recognition**: Powers virtual assistants and transcription services by converting spoken language into text.
- **Natural Language Processing (NLP)**:Enables tasks such as sentiment analysis, language translation, and chatbots through advanced models liketransformers.
- **Healthcare**: Assists in medical image analysis, predicting diseases, andpersonalizing treatment plans.

Performance & Complexity:

- DL usually provides higher accuracy on complex, non-linear tasks (e.g., computer vision), but is often a "black box" with lower interpretability compared to traditional ML methods like linear

Limitations

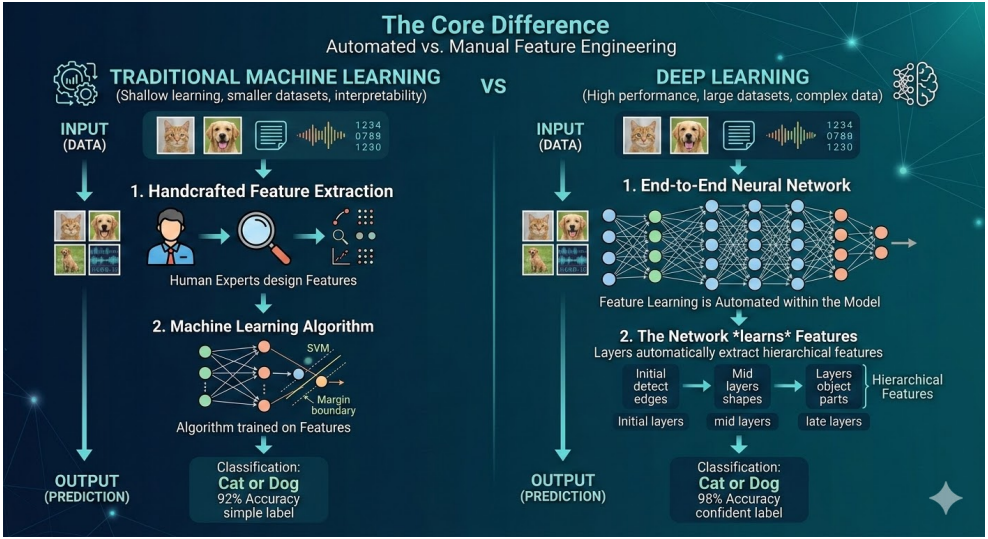
Despite its strengths, DL has several limitations

- **Data Requirements**: Deep learning models typically require vast amounts of labeled data to train effectively. This can be a challenge in domains where data collection is expensive or time-consuming.
- **Overfitting**: Insufficient data can lead to overfitting, where the model learns noise instead of meaningful patterns.
- **Computational Intensity**: Training deep learning models is computationally intensive and oftenrequires specialized hardware, such as GPUs or TPUs. This can lead to longer training times and higher costs, making deep learning less accessible for smaller organizations or projects with limited resources.

Main algorithms:

- **Convolutional Neural Networks (CNNs)**: Primarily used for image processing, focusing on spatial hierarchies and local patterns.
- **Recurrent Neural Networks (RNNs)**: Designed for sequential data, handlingtime-series data and natural language processing tasks.
- **Transformers**: A recent architecture that excels in NLP tasks by processing data in parallel and using self-attention mechanisms.

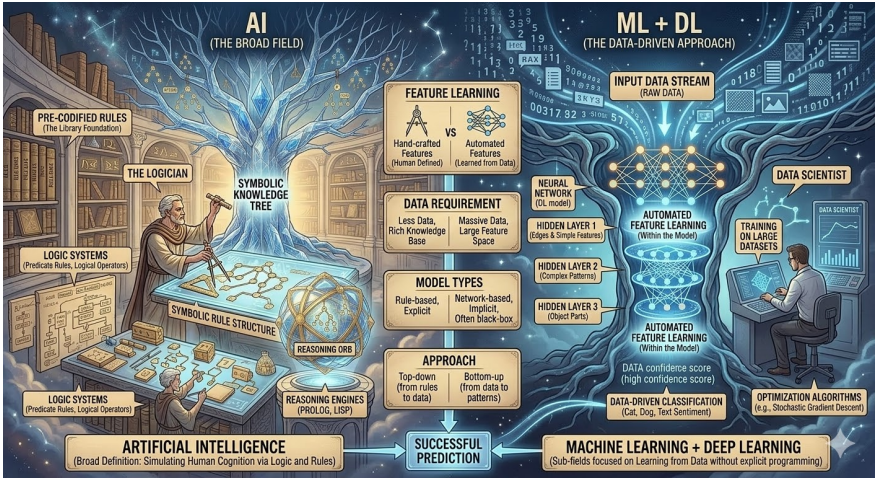
TML versus DL



Summary TML vs DL

- **Data requirements**. TML generally requires smaller datasets and can perform well with limited data if the features are carefully engineered. It is effective for structured data types. DL needs large amounts of labeled data to achieve high performance. It excels in scenarios where vast datasets are available, particularly with unstructured data like images, audio, and text
- **Manual vs. Automatic feature engineering**. TML relies heavily on manual feature engineering, requiring domain expertise to identify and select relevant features. This process can be time-consuming and may introduce bias if not done carefully. DL automatically learns feature representations from raw data through its multi-layered architecture. This capability reduces the need for manual feature extraction and allows the model to uncover complex patterns without human intervention.
- **Computational Resources**. TML typically requires less computational power and can be run on standard hardware. Training times are usually shorter, making it more accessible for smaller projects and organizations. DL demands significant computational resources, often necessitating the use of GPUs or TPUs for efficient processing. Training deep neural networks can take hours to days, to months, depending on the complexity of the model and the size of the data.
- **Explainability of Models**. TML algorithms offer greater interpretability, allowing users tounderstand how decisions are made based on input features. At the present time, this is still crucial in fields like finance and healthcare, where understanding model decisions is essential. May be it will be different in a near future.
- **Deep Learning**. Models can be challenging to interpret. Understanding how decisions aremade can be difficult due to the complexity and non-linearity of the models, which may hinder their adoption in critical applications requiring transparency.
- **Performance on Complex Problems**. TML generally struggles with unstructured data, requiring extensive preprocessing and feature extraction to be effective. Its more suited for structured data types where relationships between features can be explicitly identified.DL excels with unstructured data, such as images, audio, andtext. Its ability to automatically learn features allows it to achieve state-ofthe-art performance in tasks like image classification, natural language processing, and speech recognition.

AI vs (TML + DL)


AI
(THE BROAD FIELD)
ML + DL
(THE DATA-DRIVEN APPROACH)
PRE-CODIFIED RULES
(The Library Foundation)
THE LOGICIAN
SYMBOLIC KNOWLEDGE TREE
LOGIC SYSTEMS
(Predicate Rules, Logical Operators)
SYMBOLIC RULE STRUCTURE
REASONING ORB
LOGIC SYSTEMS
(Predicate Rules, Logical Operators)
REASONING ENGINES
(PROLOG, LISP)
FEATURE LEARNING
Hand-crafted Features
(Human Defined)
VS
Automated Features
(Learned from Data)
DATA REQUIREMENT
Less Data, Rich Knowledge Base
Massive Data, Large Feature Space
MODEL TYPES
Rule-based, Explicit
Network-based, Implicit, Often black-box
APPROACH
Top-down
(from rules to data)
Bottom-up
(from data to patterns)
INPUT DATA STREAM
(RAW DATA)
NEURAL NETWORK
(DL model)
HIDDEN LAYER 1
(Edges & Simple Features)
HIDDEN LAYER 2
(Complex Patterns)
HIDDEN LAYER 3
(Object Parts)
AUTOMATED FEATURE LEARNING
(Within the Model)
AUTOMATED FEATURE LEARNING
(Within the Model)
DATA confidence score
(high confidence score)
DATA-DRIVEN CLASSIFICATION
(Cat, Dog, Text Sentiment)
DATA SCIENTIST
TRAINING ON LARGE DATASETS
OPTIMIZATION ALGORITHMS
(e.g., Stochastic Gradient Descent)
ARTIFICIAL INTELLIGENCE
(Broad Definition: Simulating Human Cognition via Logic and Rules)
SUCCESSFUL PREDICTION
MACHINE LEARNING + DEEP LEARNING
(Sub-fields focused on Learning from Data without explicit programming)

Summary SAI vs (TML+DL)

- **Domain Knowledge**. To create effective rules, you need to understand your system deeply. If you have a good understanding of your system, then SAI will give you amazing performance. See the NetHack link I gave you.

  > *...The margin of victory was significant, with the top symbolic agent beating the top neural agent by a factor of almost 3 in the median score. This was, in fact, increased when looking at the very best agents from each team, where frequently we might see almost an order of magnitude improvement in the median score between the best symbolic and neural agents.... While our best symbolic teams had moderate-to-expert NetHack domain understanding, we were surprised to find they often had extensive ML experience as well. In fact, both winning symbolic teams said they had intended to enter the neural track, but found their symbolic methods scaled much better Symbolic bots excelled in keeping the full game state in memory, and incorporating external knowledge into their strategies. They found it easy to transfer domain knowledge to the decision-making process. In contrast, neural agents find it harder to maintain information in memory, especially if there is no reward directly associated with it.*

- Without good domain understanding, youre wasting time on trad AI. This is a drawback of AI systems since both the domain expert time to give inputs and the developer time to turn those inputs into rules is very expensive. And trying to come up with rules for very complex (or chaotic) systems is a nightmare. This is where ML can come in. Look at A Comparison of SVM against Pre-trained Language Models (PLMs) for Text Classification Tasks: the experimental results on the four datasets show that using PLMs, even fine-tuned, do not provide significant gain over the linear SVM classifier. Hence, we recommend that for text classification tasks, traditional SVM along with careful feature engineering can pro-vide a cheaper and superior performance than PLMs.

- If you have no domain understanding, or with very unstructured/complicated data then Deep Learning becomes a good friend. Vision and Language are perfect examples of this. Coming up with good features for these tasks is a nightmare (and we have heaps of data).